PYTHON 90 DAYS CHALLENGE

DAY 11 — LISTS IN PYTHON

Source coverage: Python Notes X CodeWithNishchal, pages 125–129 (Lists: creating, accessing, updating, methods and useful functions).

Day Goal

Understand how to create lists, access items using positive and negative indexes, update values, and use common list methods and built-in functions.

1. What is a List?

A **list** is a Python data structure used to store multiple values in one variable. Lists are written inside square brackets `[]`, and items are separated by commas.

Important properties

Property	Meaning	Example
Ordered	Items keep their positions	items[0]
Mutable	Items can be changed	items[1] = 'New'
Duplicates allowed	Same value can appear more than once	[10, 10, 20]
Mixed data types	Can hold different data types	['Aman', 20, True]
Indexed	Each item has an index	0, 1, 2, ...

2. Creating Lists

```
numbers = [10, 20, 30, 40]
names = ["Aman", "Riya", "Kabir"]
mixed = ["Python", 100, 99.5, True]
empty = []
nested = [[1, 2], [3, 4]]
letters = list("ABC")

print(numbers)
print(names)
print(mixed)
print(empty)
print(letters)
```

Remember: a list can contain numbers, strings, Boolean values, mixed values, or even other lists. An empty list is written as `[]`.

3. Accessing List Values

Use an index number inside square brackets to access an item. Python indexing starts from **0**. Positive indexing starts at the left; negative indexing starts from the right.

```
students = ["Aman", "Riya", "Kabir", "Neha"]

print(students[0])   # Aman
print(students[1])   # Riya
print(students[-1])  # Neha
print(students[-2])  # Kabir
```

Code	Meaning	Result
students[0]	First item	"Aman"
students[1]	Second item	"Riya"
students[-1]	Last item	"Neha"
students[-2]	Second-last item	"Kabir"

4. Updating List Values

Lists are mutable, so an existing item can be changed using its index. The index must already exist in the list.

```
marks = [70, 80, 90]
marks[1] = 85
print(marks)
# Output: [70, 85, 90]

values = [70, 80, 90, 60]
values[1] = 88
values[2] = 95
print(values)
# Output: [70, 88, 95, 60]
```

5. Important List Methods

Method	Purpose	Example
append(x)	Adds one item at the end	items.append(40)
extend(iterable)	Adds multiple items	items.extend([50, 60])
insert(i, x)	Adds an item at a specific index	items.insert(1, 15)
remove(x)	Removes the first matching value	items.remove(20)
pop(i)	Removes and returns item at index i	items.pop(1)
clear()	Removes all items	items.clear()
index(x)	Returns the index of a value	items.index(30)
count(x)	Counts occurrences of a value	items.count(10)
sort()	Sorts the list in ascending order by default	items.sort()
reverse()	Reverses the list order	items.reverse()
copy()	Creates a shallow copy	new = items.copy()

Example using methods

```
numbers = [10, 20, 30]
numbers.append(40)
numbers.insert(1, 15)
numbers.remove(20)
print(numbers)
# Output: [10, 15, 30, 40]
```

6. Useful Built-in Functions with Lists

Function	Purpose	Example
len(list)	Counts the number of items	len(numbers)
sum(list)	Adds numeric items	sum(numbers)

Function	Purpose	Example
min(list)	Returns the smallest value	min(numbers)
max(list)	Returns the largest value	max(numbers)
sorted(list)	Returns a sorted copy; original list is unchanged	sorted(numbers)

7. Coding Practice — Try Without Solutions

Basic

1. Create a list containing five numbers and print it.
2. Create a list of three student names and print the first name.
3. Create a mixed list containing a string, an integer, a float and a Boolean value.
4. Create an empty list and print it.

Intermediate

5. Given students = ['Aman', 'Riya', 'Kabir', 'Neha'], print the first, last and second-last items.
6. Given marks = [70, 80, 90], change the second item to 85 and print the list.
7. Start with numbers = [10, 20, 30]. Append 40, insert 15 at index 1, remove 20, then print the result.
8. Create a list with repeated values and use count() to count how many times one value appears.

Placement Practice

9. Take a list of five numbers and display its length, sum, minimum and maximum.
10. Sort a list using sorted() and show that the original list remains unchanged.
11. Create a list of numbers, reverse it using reverse(), and print the result.
12. Use copy() to create a copy of a list, update one list, and observe whether the other list changes.

8. Interview Questions

1. What is a list in Python?
2. How do you create an empty list?
3. What does it mean that a list is mutable?
4. What is the difference between positive and negative indexing?
5. What is the difference between append() and extend()?
6. What is the difference between remove() and pop()?
7. What does clear() do?
8. What is the purpose of index() and count()?
9. What is the difference between sort() and sorted()?
10. What is the purpose of len(), sum(), min() and max() when used with lists?

Day 11 Completion Checklist

■ I can create lists with different data types.

■ I understand positive and negative indexing.

■ I can update list items using indexes.

■ I can explain and use the list methods covered today.

■ I can solve the practice questions without copying solutions.